Mahmoud Alyosify

MLOps & Cloud AI Engineer | Model Deployment, Pipelines & Cost-Efficient Inference
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PROFESSIONAL SUMMARY

MLOps & Cloud AI Engineer focused on the gap between a trained model and a system that runs reliably and affordably. Currently an **AI Engineer Intern at Injaz Tech**, taking an agentic RAG system to a **FastAPI** service with **Docker**, **GitHub Actions** CI, an automated evaluation gate, Prometheus metrics and per-node cost reporting. Deployed **fine-tuned LLMs and vision encoders** to **AWS** via **ONNX, Docker and FastAPI**; built **deterministic, reproducible benchmarking on pinned GPU instances** measuring **TTFT, inter-token latency and GPU energy through NVML**; and trained **deep-RL agents that provision cloud resources against an explicit cost–latency budget**. Holds **three AWS Academy credentials** (ML Foundations, ML for NLP, Data Engineering). Completing an **MSc in Artificial Intelligence at Queen's University, Canada** (expected **Nov 2026**) on a **Digilians Presidential Scholarship**, with prior production-systems experience on **national-scale toll infrastructure**.

EDUCATION

MSc in Artificial Intelligence — *Sep 2025 – Nov 2026*
Queen's University, Kingston, Ontario, Canada — Expected Nov 2026 · GPA: 4.30 / 4.30

- **Digilians Presidential Scholarship** — competitive national AI graduate fellowship.
- Coursework: **Cloud Computing, End-to-End ML Pipelines, Deep Learning, Large Language Models, Reinforcement Learning**.

B.Sc. Computer & Information Science (Bioinformatics) — *Sep 2019 – Jul 2023*
Assiut University, Egypt — GPA: 3.53 / 4.00

- Graduation Project: **Vitalism Solution** — **Grade A+**; **Semi-Finalist, Microsoft Imagine Cup 2023**.

TECHNICAL SKILLS

Cloud & Infrastructure	AWS (GPU Instances, ML & Data Services), Docker, Docker Compose, CI/CD (GitHub Actions), Linux, Bash, Infrastructure Cost Optimisation, Autoscaling Strategies, Self-Hosted Model Serving
Model Serving & Deployment	ONNX, FastAPI, Streamlit, FAISS, ChromaDB, Containerised Inference, Batch & Interactive Serving, SSE Streaming, Health & Readiness Probes, Response Caching, API Design
Reliability & Measurement	Deterministic & Reproducible Pipelines, Pinned Environments and Seeds, Latency Profiling (TTFT, Inter-Token), Throughput Benchmarking, GPU Energy Monitoring (NVML), Baseline Comparison, Retries & Circuit Breakers, Structured Logging, Prometheus Metrics, CI Evaluation Gates, pytest
Data Engineering	ETL Pipelines, PySpark, Pandas, NumPy, SQL, Data Validation, MySQL, MongoDB, SQL Server
ML & AI	PyTorch, TensorFlow, Scikit-learn, Hugging Face, LangChain, LangGraph, RAG, Fine-Tuning (LoRA, QLoRA, PEFT), Reinforcement Learning (PPO, DQN), Optuna
Programming	Python, SQL, C#, C++, C, Go, JavaScript, Bash

PROFESSIONAL EXPERIENCE

AI Engineer Intern — *Oct 2026 – Present*
Injaz Tech — Injaz Group (formerly Fingerprint Consultancy) — Maadi, Cairo, Egypt

- Took an agentic RAG assistant (LangChain, LangGraph, FAISS) from notebook to service: **FastAPI** with SSE streaming and liveness/readiness probes, typed **pydantic-settings** configuration, and a multi-stage **Docker** / docker-compose setup with a persisted index volume.
- Built a **GitHub Actions** pipeline (lint plus **136** offline tests on every PR, and a nightly **evaluation gate** that fails the build when quality drops below a committed baseline), with structured JSON logging, **Prometheus** metrics and per-node token and cost reporting.
- Hardened it with timeouts, exponential-backoff retries, a circuit breaker, a hard per-request LLM-call budget, rate limiting and an answer cache that cut repeat-question latency from **34.7 s to 1.6 s**.

Part-Time Instructor — Machine Learning & AI for Business — *Jul 2025 – Present*
National Telecommunication Institute (NTI) — Cairo, Arish & Assiut, Egypt

- Teach **Machine Learning** (90-hour bootcamps in Cairo, Arish and Assiut) and **AI for Business** modules in NTI's Summer Internships Program, on separate engagements.

AI in Cybersecurity Intern — *Jun 2025 – Jul 2025*
e& Egypt — via NTI (On-the-Job Training) — Cairo, Egypt

- Applied AI-driven security tooling across threat intelligence, digital forensics and enterprise security analysis.

Systems Engineer (Military Service) *Feb 2024 – Mar 2025*
National Company for Roads Building & Development Cairo, Egypt

- Supported production backend systems for national toll infrastructure serving **50%+ of Egyptian vehicles** across **70%+ of national roads**, where downtime had immediate operational cost.
- Debugged and optimised real-time transportation software and supported legacy applications to maintain continuity; co-developed an internal data-management and reporting application in **C#, .NET, Entity Framework, SQL Server** and **Crystal Reports**.

Online Instructor — AI, Algorithms & Data Science *2020 – Present*
Udemy & YouTube Remote

- Built Arabic-language technical curricula reaching **10,000+ learners** and **700K+ YouTube views**.

SELECTED DEPLOYMENT & INFRASTRUCTURE PROJECTS

RL-Cloud-Autoscaler — Learned Cloud Resource Provisioning *2026*
Python · PPO · DQN · Gymnasium · Cloud Cost Optimisation

- Built a **custom Gymnasium environment** simulating cloud workload dynamics, then trained **PPO** and **DQN** agents to make provisioning decisions in real time.
- Optimised an explicit **cost–latency reward** and benchmarked against **static allocation** and **reactive threshold-based autoscaling** — the policies teams actually run — to quantify real savings rather than simulated ones.

MinimalLM: Verbosity-Aware Decoding for Efficient Language Model Generation *2026 – Present*
MSc Research Project (CISC 898), Queen's University · NVIDIA A40 GPU · NVML

- Training-free decoding-time control of LLM verbosity, benchmarked with **deterministic, reproducible runs** (fixed seeds, versioned environments, pinned **NVIDIA A40** GPU) and **TTFT, inter-token latency and GPU-energy** measurement on three model families. An entropy-gated logits processor cut **25.5–34.9%** of MT-Bench tokens within pre-registered margins, but an audit showed the saving was truncation.
- A survey of **23** training-free mechanisms found none that beat truncation without oracle labels; an **NLI-licensed elision** method beats it on every model but removes only 0.9–2.2% of tokens.

Horus-OSINT — Fine-Tuned LLM Deployed on AWS *2026*
Meta-Llama-3-8B · Fine-Tuning · AWS · Self-Hosted Serving

- Owned the full path from a **159,826-record** dataset through fine-tuning to a running **AWS-hosted** interactive service — data preparation, training, packaging and deployment.

SimCLR-Vision-SSL — ONNX Export & Vector-Search Serving *2026*
PyTorch · ONNX · FAISS · Retrieval

- Exported a trained encoder to **ONNX** and served embeddings through a **FAISS** index — converting a research model into a framework-independent, deployable retrieval component.

Additional Projects

- **HORUS Sentinel** — multi-agent platform with a self-hosted fine-tuned LLM and knowledge-graph state layer running continuously.
- **Automated Multimodal Agent** — multi-stage automation pipeline chaining extraction, generation and synthesis services.
- **Also:** SCRAP · AudioShield (Streamlit-served CNN) · Human vs. Agentic Pull Requests · Vitalism Solution.

CERTIFICATIONS & AWARDS

AWS Academy — Data Engineering
Machine Learning Specialization — DeepLearning.AI
Full Stack Diploma — .NET & Angular (Route Academy)
1st Place — Smart Cities Hackathon 2022
AWS Academy — ML Foundations, ML for NLP
HCIP – AI (Huawei)
Microsoft Certified: Data Analyst Associate
Semi-Finalist — Microsoft Imagine Cup 2023

LANGUAGES

- **Arabic:** Native · **English:** Professional Working Proficiency · **French:** Basic.